

Spatio-temporal evolution of resilience: The case of the Chengdu-Chongqing urban agglomeration in China

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ABSTRACT

Cities, as complex systems, are increasingly vulnerable to a range of natural hazards and public safety risks, exacerbated by the rapid pace of global urbanization. Understanding these complex urban systems and developing sustainable development plans is crucial for policymakers, necessitating a precise evaluation of urban resilience. This study presents an innovative, data-driven approach to analyze urban resilience, integrating the Functional Resonance Analysis Method (FRAM), Modified K-shell (MKS) decomposition algorithm, and Bayesian Networks (BNs). Focusing on the Chengdu-Chongqing urban agglomeration, we utilize spatio-temporal analysis to uncover resilience evolution and patterns. The findings indicate: (1) The distribution of city resilience in the Chengdu-Chongqing urban agglomeration exhibits a remarkable two-headed structure; (2) a diminishing resilience variance among urban agglomerations over time, suggesting an α -convergence trend; (3) a negative correlation in spatial correlation analysis for the Chengdu-Chongqing urban agglomeration; (4) After considering the spatial correlation, the Chengdu-Chongqing urban agglomeration shows an obvious β -convergence trend. These insights suggest urban planners should prioritize equitable resource distribution and adopt synergistic and integrated development strategies to mitigate negative spillovers and ensure balanced growth, aiding the development of core cities in a way that benefits their surrounding areas.

1. Introduction

Global urbanization is a transformative trend in human society in the 21st century (Korhonen & Snäkin, 2015). As early as 1996, the United Nations Human Settlements Programme declared humanity's entry into the "urban era," marked by the urban population surpassing the rural population for the first time. By 2030, it is anticipated that nearly 5 billion people will reside in urban areas, accounting for almost 75 % of global carbon emissions and energy consumption (Kennedy et al., 2014). Cities, as complex systems encompassing social, economic, ecological, and infrastructural elements, face diverse risks (Fan et al., 2017). Urbanization and industrialization, while beneficial for human well-being, have exacerbated these risks, leading to challenges like geological and hydrological disasters, public health crises, ecological degradation, and resource overconsumption (Shi et al., 2021). In light of these challenges, urban resilience has emerged as a novel concept in urban risk governance and a key topic in international urban planning. Urban resilience is broadly defined as the ability of urban systems to resist, adapt to

threats, and recover to their original state after disruption. Studies have indicated that cities with high resilience can effectively and promptly respond to disasters, significantly reducing losses in life, economy, and environment caused by such risks (Birhanu et al., 2017). Currently, Chinese cities are in a critical phase of formulating their 14th Five-Year Plan. It is crucial to incorporate resilience into the overall development strategy during urban planning in areas such as transportation, construction, and industry (Huang et al., 2021). This requires not only a deep understanding of the importance and practical level of urban resilience by planners and managers, but also highlights the necessity of using scientific and advanced technology for urban planning, with the aim of building cities that are both smart and highly resilient (Huang et al., 2022).

In recent decades, China has experienced rapid urbanization, with its urbanization rate reaching 59.58 % by the end of 2018 and expected to hit 70 % by 2030, according to the country's economic plans (Dong et al., 2020). Urban agglomerations have become the predominant spatial structure in rapidly urbanizing regions (Zhao, Wang, & Wang,

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2021), with notable examples such as the Jing-Jin-Ji in the north, Yangtze River Delta in the east, Pearl River Delta in the south, and Chengdu-Chongqing in the west. In these urban clusters, closely intertwined and complex city relationships have escalated risks such as geological hazards (Lin et al., 2021) and environmental issues (Cui et al., 2021). Ensuring high resilience in these agglomerations is imperative for sustainable urban development, necessitating precise assessments of resilience to guide policymaking and strategy formulation. However, research on resilience in urban agglomerations is still nascent, with most studies focusing on individual city resilience and limited exploration of varying resilience levels across cities within an urban cluster (Ribeiro & Gonçalves, 2019). This study introduces an innovative data-driven method for quantifying urban resilience, integrating Functional Resonance Analysis Method (FRAM), modified K-shell (MKS) decomposition algorithm and Bayesian Networks (BNs) algorithm. The Chengdu-Chongqing urban cluster serves as a case study to demonstrate this model's application. Additionally, this research employs spatial-temporal analysis to discern the evolving resilience patterns within the Chengdu-Chongqing urban cluster, providing insights crucial for developing urban resilience strategies.

The structure of this study is as follows. Section 2 provides a literature review and shows the innovations in this study. Section 3 introduces the evaluation framework of urban resilience from the perspective of complex adaptive systems. Section 4 presents the application of the proposed methodology. Section 5 conducts case studies, followed by analysis and discussion of the results. Finally, Section 6 concludes the study and proposes future work.

2. Literature review

2.1. Theoretical framework for urban resilience

The concept of resilience, originally proposed by Holling (1973) to describe the capacity of ecosystems to withstand changes in environmental variables, has since been expanded to various systems. These include organizational (Barasa et al., 2018), physical (Kong et al., 2021), social (Alizadeh & Sharifi, 2021), and socio-technical systems (Caputo et al., 2020). Despite the absence of a universally accepted definition (Qiao et al., 2021), there is a consensus that resilience involves a system's ability to absorb damage, adapt to changing environments, and recover from disruptions (Gao et al., 2016). Cities, as complex systems, are composed of interlinked economic, social, and natural elements. They face diverse risks including natural disasters (earthquakes, floods, sandstorms), ecological and environmental threats (pollution, ecological destruction), and public health crises (pandemic outbreaks). To address these challenges, the International Council for Local Environmental Initiatives (ICLEI) introduced the concept of "resilient cities" in 2002, focusing on urban disaster prevention. The interpretation of urban resilience and risk resistance, however, varies across disciplines, resulting in distinct theoretical frameworks. These frameworks are focused on three main aspects: (1) The complex adaptive system viewpoint, which emphasizes the dynamic evolution of urban systems in response to disturbances. This perspective is supported by Holling and Gunderson's (2002) adaptive cycle model, viewing resilience as a state of continuous dynamic changes through a systems theory lens. (2) The risk and disaster perspective, which concentrates on the sustainability of urban systems and emphasizes disaster risk governance (Manyena, 2006). This view is exemplified by Cutter et al.'s (2008) Disaster Resilience of Place Model (DROP), accounting for pre-disaster community conditions, disaster characteristics, and absorption capacity, and is applicable to both rapid-onset and slow, cumulative impact disasters. (3) The urban community perspective, which considers communities as essential components of cities. Communities are often the first affected by disasters and disturbances and are crucial for city resilience (Reed et al., 2009). Francis and Bekera's (2014) Resilience Matrix Framework (RM) provide a comprehensive tool for assessing and improving

community-level resilience, incorporating various aspects of urban life.

2.2. Methodological approaches in urban resilience

Evaluating urban resilience is a critical component in resilience research. Understanding the time and process involved in a city's recovery to its pre-disturbance state is essential. More crucial, however, is the ability to accurately predict urban resilience and proactively prepare for potential disturbances. On one hand, assessing urban resilience enhances our comprehension of its structural makeup. On the other hand, it facilitates informed planning based on a city's resilience capabilities. Despite the lack of a standardized approach in academia for assessing urban resilience, employing indicators related to urban resilience offers a scientific method for its quantification. Common methods for quantifying urban resilience include comprehensive index methods such as expert scoring (Qasim et al., 2016), analytic hierarchy process (Zhang et al., 2020), principal component analysis (Ebisudani & Tokai, 2016), the entropy method (Zhao, Fang, et al., 2021), TOPSIS (Zhu et al., 2019), and the CRITIC weighting method (Liu, Lei, et al., 2022). However, many of these methods rely on subjective judgment, presenting challenges in objective data analysis. Given that a city functions as a complex system, its resilience factors are inherently non-linear (Huang et al., 2021), and traditional quantitative methods fail to capture the interactive mechanisms between different systems (Liu, Ma, & Wang, 2022). Additionally, most urban resilience studies are limited to simple calculations and assessments, often overlooking the dynamic evolutionary trends and convergence (Lu, Lu, et al., 2022). Bayesian Networks (BNs) are a type of machine learning method based on the probabilistic graphical models of probability theory and mathematical statistics. Each BN comprises a Directed Acyclic Graph (DAG) and a Conditional Probability Table (CPT), which together capture the relationships between variables on both qualitative and quantitative levels. Qualitatively, the DAG delineates the influence relationships among variables, offering a visual representation of their dependencies and impacts within a system. Quantitatively, the CPT details the joint probability distribution of these variables. BNs have found extensive application in resilience assessment across various fields, providing valuable insights into the evaluation of resilient cities. For example, Tang et al. (2020) introduced a hierarchical BN Model specifically designed to quantitatively assess the resilience of urban agglomeration transportation systems. Hosseini and Barker (2016) utilized BNs to explore resilience-building strategies for infrastructure systems. Similarly, Hossain et al. (2019) employed BNs to identify and summarize the key factors that enhance the resilience of port infrastructure systems.

2.3. Gaps and innovations

The literature review allows us to draw several important conclusions: (1) Presently, urban resilience assessment methods are predominantly subjective and lack incorporation of quantitative methods and data. This heavy reliance on subjective judgments could potentially undermine the objectivity and reliability of the assessment outcomes. (2) Most evaluation methodologies give limited attention to understanding the underlying mechanisms of urban resilience and conducting multi-scale evaluations. They tend to neglect the interaction among various resilience elements and focus predominantly on single-scale assessment results. (3) The majority of existing literature views the resilience of individual cities from a static standpoint, with insufficient focus on the spatiotemporal dynamics of resilience in urban agglomerations. This approach often misses capturing the evolving trends of urban resilience over time and across different spaces. To address these shortcomings, this paper develops a research framework and an evaluation index system for urban resilience, drawing from the perspective of complex adaptive systems. It proposes a data-driven, quantitative evaluation method for resilient cities, utilizing the Functional Resonance Analysis Method (FRAM), Modified K-shell (MKS) decomposition

algorithm, and Bayesian Network (BN). Moreover, this study delves into the spatial and temporal evolution of urban agglomerations' resilience through the use of spatial autocorrelation analysis, and convergence analysis. The innovations and contributions are as follows:

- Our study introduces a novel multi-level indicator evaluation system inspired by the theory of complex adaptive systems; Under this framework, we conducted a systematic review to select indicators from five key dimensions: environment, economy, society, infrastructure, and government. This rigorous selection process was designed to minimize bias and ensure the indicators' transparency and objectivity.
- We have developed an innovative, data-driven approach to quantify urban resilience. This method integrates the FRAM, MKS decomposition algorithm, and BN, effectively overcoming the potential biases of expert-driven methods and eliminating the quantitative limitations in FRAM. The applicability and effectiveness of this approach were verified through its application to the Chengdu-Chongqing urban agglomeration in China.

3. Framework for urban resilience evaluation

3.1. Application of complex adaptive systems theory in urban resilience

Complex Adaptive Systems (CAS) theory critically examines the self-organization mechanisms of systems, suggesting that agents within these systems display dynamic behavior and undergo adaptation and learning processes (Basu et al., 2019). Urban systems, as instances of CAS, intricately intertwine human and natural elements. They are characterized by a continuous cycle of evolution towards either prosperity or decline. This evolutionary cycle, marked by oscillations and the emergence of new forms, is driven by the interactions of material, energy, and information flows within and beyond the urban environment, across diverse spatial and temporal scales. Such dynamics are indicative of a city's ongoing evolution, guided by self-organization and proactive learning as it adapts to environmental changes.

The study of urban resilience is an important dimension for further understanding the complex evolutionary processes of urban systems. CAS demonstrate change characteristics, whether or not they face external disturbances. The resilience of these systems is characterized not by a return to a pre-disturbance state, but by their capacity for change, adaptation, and transformation under stress, showcasing unique evolutionary features (Carpenter et al., 2005). Urban resilience encompasses a spectrum of capabilities that enable an urban system to adaptively develop under stress. This resilience is generally comprised of three key capabilities (Fang et al., 2017; Davoudi et al., 2016): (1) The ability to maintain normal functions independently during emergencies such as earthquakes, floods, and public health crises; (2) The capacity to restore normalcy when subsystems are disrupted; and (3) The ability to employ structural adjustments and learning experiences to confront future challenges. These capabilities are mirrored in four distinct characteristics of urban systems: (1) Stability, serving as a foundational mechanism, signifies the system's resilience to disturbances and its

ability to maintain the city's core structure; (2) Self-organization, indicative of the dynamic supplementary mechanisms inherent in a CAS, enhances the city's capacity to manage impacts; (3) Learning adaptability, highlighting the system's innovative responses to environmental shocks; and (4) Transformability, the ability to regain normal functioning and achieve a new equilibrium post-impact. In light of this, from the CAS resilience perspective and considering the five subsystems of the city (environment, economy, society, infrastructure, and government), this paper proposes an open, multidimensional, and multi-factor coupled urban resilience evaluation framework as illustrated in Fig. 1. This framework aspires to be a comprehensive tool for the systematic exploration of urban resilience.

(1) Stability: Economic resilience minimizes the economic losses caused by external disturbances, maintaining the continuity of urban economic activities, reducing the risk of unemployment and economic recession, thereby sustaining urban stability and public confidence (Du et al., 2019). Social resilience enhances the city's inherent strength and external response capability when facing shocks, helping to reduce conflicts and social unrest (Hosseini et al., 2020). Additionally, strong infrastructure resilience ensures that the city can maintain critical services and functions in the event of disasters, protecting the stability of the city's physical dimensions (Haggag et al., 2022).

(2) Self-organization: Cities with high social resilience can spontaneously generate strategies to confront challenges through tight social networks, high levels of trust, and a spirit of cooperation (Anzola et al., 2017). Economic resilience promotes self-adjustment and innovation, enabling individuals, businesses, and organizations to adapt to external changes and independently form new cooperation modes and solutions without central guidance (Sharifi, 2021).

(3) Learning adaptability: Government resilience is reflected in the recovery and adaptability of institutional frameworks and governance systems under pressure, based on learning from historical experiences and flexibly adjusting policies, urban planning, and management decisions (Li, Huang, et al., 2021; Huang et al., 2023). Environmental resilience emphasizes learning from environmental changes, promoting continuous improvement in urban planning, design, and management, reflecting the city's learning and progress in environmental adaptability (Kuang & Liao, 2020; Li et al., 2020).

(4) Transformability: Economic resilience enables the city to readjust its economic development path, reorganize its economic structure, and seize new opportunities to meet challenges after suffering shocks (Yang et al., 2021). Infrastructure resilience ensures that the city's physical systems can continue to operate in the face of shocks, supporting the recovery and transformation of the socio-economic structure (Hu & Hassink, 2020). Government resilience show that the city can promptly adjust rules, processes, or structures when faced with shocks, ensuring it can adapt to new situations and turn towards more favorable development directions (Giulia, 2023).

3.2. Selection of urban resilience indicators

3.2.1. Methodological overview

Systematic reviews, originating in the medical field, systematically

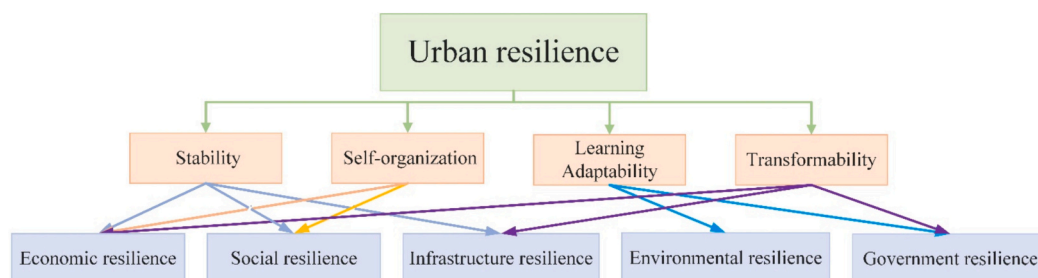


Fig. 1. Evaluation framework for urban resilience from the perspective of complex adaptive system.

compile relevant research findings using evidence-based medicine and clinical epidemiology principles. This approach involves rigorous literature evaluation and synthesis of studies that meet specific criteria to form comprehensive conclusions (Pickering & Byrne, 2014). This study adopts the systematic review method, focusing on analyzing literature about urban resilience measurement indicators. We establish strict criteria to minimize selection bias, enhancing indicator identification's consistency and objectivity. The PRISMA framework guides our literature selection and data extraction, ensuring thoroughness and reproducibility (Godschalk, 2003).

3.2.2. Literature retrieval and screening

The academic consensus is that the concept of “resilience” began permeating urban management and planning disciplines in the late 1990s (Moher et al., 2010). In line with this, our paper commences the literature search from 1990, methodically retrieving international publications on urban resilience measurement indicators up until 2023. To encompass the multidisciplinary nature of urban resilience, a range of databases were selected for the search, including Scopus, Web of Science, PubMed, Science Direct, and CNKI, complemented by cross-verification using Google Scholar. For keyword selection, aligned with the research theme, a series of terms such as “resilience/resilient,” “tool*,” “measure*,” “indicator*/index,” “assess*/survey,” “framework/model,” “design,” and “city/urban/megacity” were utilized to explore relevant topics in titles, keywords, and abstracts. To ensure an unbiased approach in the search process, two graduate students independently conducted the searches and cross-checked the results.

A defining feature of a systematic review is the establishment and rigorous adherence to specific literature inclusion and exclusion criteria, tailored to the research theme. The inclusion criteria for this study are as follows:

- (1) Literature that centers on the definition, characteristics, and measurement indicators of urban resilience, particularly focusing on the measurement indicators.
- (2) Articles published in English in peer-reviewed journals and original research articles.
- (3) Publications dated between January 1990 and July 2023.
- (4) Studies where the unit of analysis is the city, rather than individuals or households.

Conversely, the exclusion criteria include:

- (1) Papers that merely mention “resilience” without it being the central focus.
- (2) Research focusing primarily on hazards, vulnerabilities, impact, or risk assessments.
- (3) Non-research publications, such as book reviews, conference papers, and book chapters.
- (4) Studies assessing a specific aspect of urban resilience, for example, flood resistance.
- (5) Research evaluating only single scales or indicators.

Following these criteria and employing the PRISMA selection process, as shown in Fig. 2, a total of 89 articles were ultimately included in this study.

Prior to coding the entire set of articles, two graduate students independently coded a sample of the articles to establish inter-coder reliability. The consistency between the coders was quantified using Cohen's Kappa value (Landis & Koch, 1977). The coding process in this study demonstrated excellent consistency, with Kappa values for each variable ranging between 0.82 and 0.90. Following this validation, all articles were coded independently. Any discrepancies in coding were thoroughly discussed and reconciled to ensure a consensus among the authors regarding the coding outcomes.

3.2.3. Determination of index system

According to the urban resilience assessment framework defined in Section 3.1, five primary indicators (society, economy, infrastructure, environment and government) were applied to the coding of 89 articles.

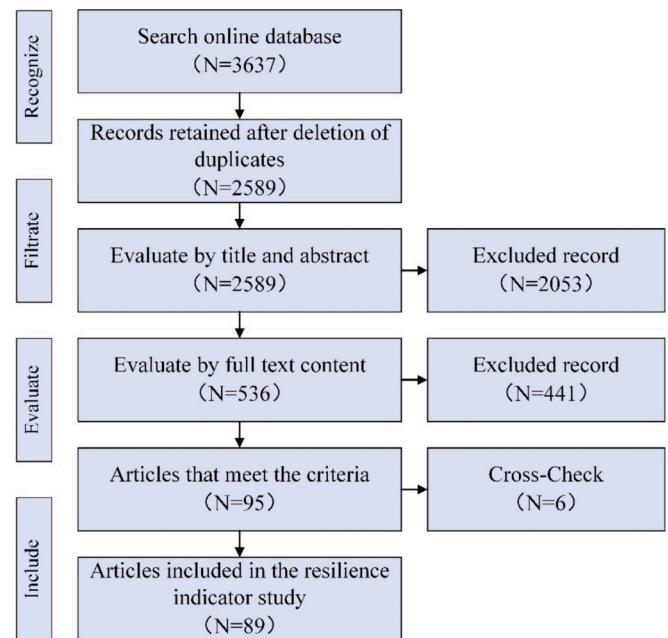


Fig. 2. Literature retrieval and screening process.

The results, as shown in Table 1, comprised a compilation that included the five primary indicators and 25 evaluation indicators. Among the 52 studies that explicitly utilized the urban resilience definition, 19 articles (36.5 %) incorporated all five primary indicators, 7 articles (13.5 %) included four, and the remaining 26 articles (50.0 %) incorporated three or fewer primary indicators.

4. Research methodology

Fig. 3 illustrates a comprehensive approach to analyzing the resilience of urban agglomerations. First, the study explores the interconnections and influence mechanisms among resilience indicators via the FRAM, leading to the creation of a specific FRAM model. The FRAM model is then transformed into a complex network to facilitate topological analysis. Employing the MKS decomposition algorithm, the weighted degree of the indicators is computed, determining their influence intensity, which lays the foundation for establishing the conditional distribution of the BN model. Following this, the BN model, incorporating historical data, is utilized to assess urban resilience. Finally, the spatial autocorrelation analysis and convergence model are used to analyze the spatio-temporal evolution of urban agglomeration.

4.1. Functional resonance analysis method

As discussed in Section 3, cities constitute complex adaptive systems in which numerous factors and functions influencing urban resilience interact and depend on each other in various contexts. Traditional linear models often oversimplify the interactions among these resilience factors, failing to capture their dynamic relationships, which results in limited resilience assessment outcomes. Hence, we have adopted the Functional Resonance Analysis Method (FRAM) to conduct a more in-depth analysis of urban resilience. FRAM, a function-based systems approach, is utilized to investigate safety-related issues and challenges within complex systems (Hollnagel, 2017). Unlike traditional approaches that emphasize physical aspects, FRAM focuses on functional aspects, dynamic interactions, and performance variability to explain how these systems operate (Sujan et al., 2020). To optimize functionality and output, FRAM employs four fundamental principles, explained as follows (Guo et al., 2024):

Table 1
Urban resilience measurement indicator system.

Target indicator	Primary indicators	Evaluation indicators	Index	Unit	Description	Source
Urban resilience	Society	Social Security and Employment Expenditures	F1	billion yuan	Measuring the degree of perfection of social security	Wang et al., 2018
		Number of College Students Enrolled	F2	10,000 persons	Reflecting the potential of the social system	Liu, Lei, et al., 2022
		Population Density	F3	person/km ²	Reflecting a city's disaster resilience	Zhao, Fang, et al., 2021
		Urbanization Rate of the Population	F4	%	Characterizing a city's disaster response capability	Nathwani et al., 2019
		Natural Population Growth Rate	F5	%	Reflecting the city's population carrying capacity	Lu, Zhang, et al., 2022
	Economy	GDP Per Capita	F6	10,000 yuan	Reflecting the development level of the urban economy	Lu, Lu, et al., 2022
		Proportion of Tertiary Industry in GDP	F7	%	Reflecting the rationality of the urban economic structure	Li, Kou, and Wen, 2021
		Per Capita Disposable Income of Urban Residents	F8	10,000 yuan	Revealing the economic vitality of the city and the quality of life of the population	Lu, Lu, et al., 2022
		Urban Fiscal Revenue	F9	billion yuan	Representing the payment capability for urban emergency rescue	Woolf et al., 2016
		Number of New Jobs	F10	10,000 persons	Reflecting the stability and growth potential of the urban economy	Cutter et al., 2014
		Total Import Value of Foreign Trade	F11	billion dollars	Measuring the external economic dependency of the city	Burton, 2015
	Infrastructure	Per Capita Area of Roads	F12	m ² / person	Reflecting the rationality of urban road resource allocation	Lu, Lu, et al., 2022
		Density of the Road Network	F13	km/ 100 km ²	Reflecting the response capacity of urban infrastructure after disasters	Gaber et al., 2019
		Number of Medical Institutions	F14	units	Measuring the urban emergency rescue and medical response capability	Liu, Ma, and Wang, 2022
		Average Daily Electricity Consumption Per Capita	F15	kWh/ person	Reflecting the completeness of infrastructure and power supply stability	Cheng et al., 2023
		Average Daily Water Consumption Per Capita	F16	L/person	Representing water supply stability and the ability to prevent post-disaster disease spread	Wang and Yan, 2023
	Environment	Greening Rate of the Built-up Area	F17	%	Describing the ecological self-purification capacity of the city	Lu, Lu, et al., 2022
		PM2.5 Concentration	F18	µg/m ³	Reflecting the degree of air pollution in the city	Jiao et al., 2023
		Rate of Good Air Quality	F19	%	Representing the overall air quality of the city	Wang et al., 2018
		Proportion of Surface Water Sections with Good Quality	F20	%	Measuring the quality and availability of urban water resources	Lu, Lu, et al., 2022
	Government	Number of Hazardous Houses Inspected	F21	units	Reflecting the institutional preparedness for disaster prevention	Xu et al., 2019
		Number of Community Public Service Stations	F22	units	Describing the distribution and accessibility of municipal resources	Shi et al., 2021
		Number of Letters Received by the Government from the Public	F23	10,000 units	Representing the effectiveness of municipal services and public satisfaction	Ghouchani et al., 2021
		Density of Government Institutions	F24	unit/km ²	Reflecting the coordination capability of the organizational system	Cheng et al., 2023
		Proportion of Heads in Government Institutions	F25	%	Reflecting the degree of participation of organizational management personnel	Cheng et al., 2023

(1) Equivalence of Successes and Failures: Both success and failure manifest through similar processes. The same factors may contribute to either favorable outcomes or accidents, with identical mechanisms influencing the end results.

(2) Approximate Adjustment: Task accomplishment, involving executing system functions and fulfilling system requirements, necessitates making suitable adjustments under prevailing conditions (time, manpower, information, resources, etc.). These adjustments are typically approximate rather than exact.

(3) Emergent Outcomes: Systemic accidents often result from incremental changes, emerging due to the variability inherent in daily adjustments, rather than from a confluence of multiple forces.

(4) Functional Resonance: Minor variability in interconnected tasks or functions can interact and intensify, leading to increased overall system variability.

In FRAM modeling, the first step is the identification and definition of functions, as delineated in Fig. 4. Within a typical urban complex system, these functions result from the synergy among technologies, humans, and organizations. They can be categorized based on six dimensions:

(1) Input: What is required to execute this function? (Inputs connect with preceding functions and may be transformed or utilized by the function to produce outputs.)

(2) Output: What does the function yield? (Outputs encompass, but are not limited to, alterations in information, sequence, and parameters.)

(3) Precondition: What prerequisites must the system fulfill before executing a function? (This includes other processes, steps, and specific conditions.)

(4) Resource: What consumable resources are essential for this function? (including human resources, equipment, electricity, and fuel.)

(5) Control: What are the constraints on performing the function? (including guidelines, plans, and procedures.)

(6) Time: What are the specific temporal requirements for this function?

The second step in developing the FRAM model is to identify the potential or actual variability of the function, as illustrated in Fig. 4, with a preference for actual instances. In urban complex systems, variability principally originates from technological, human, and organizational functions (Li et al., 2019). Understanding functional variability

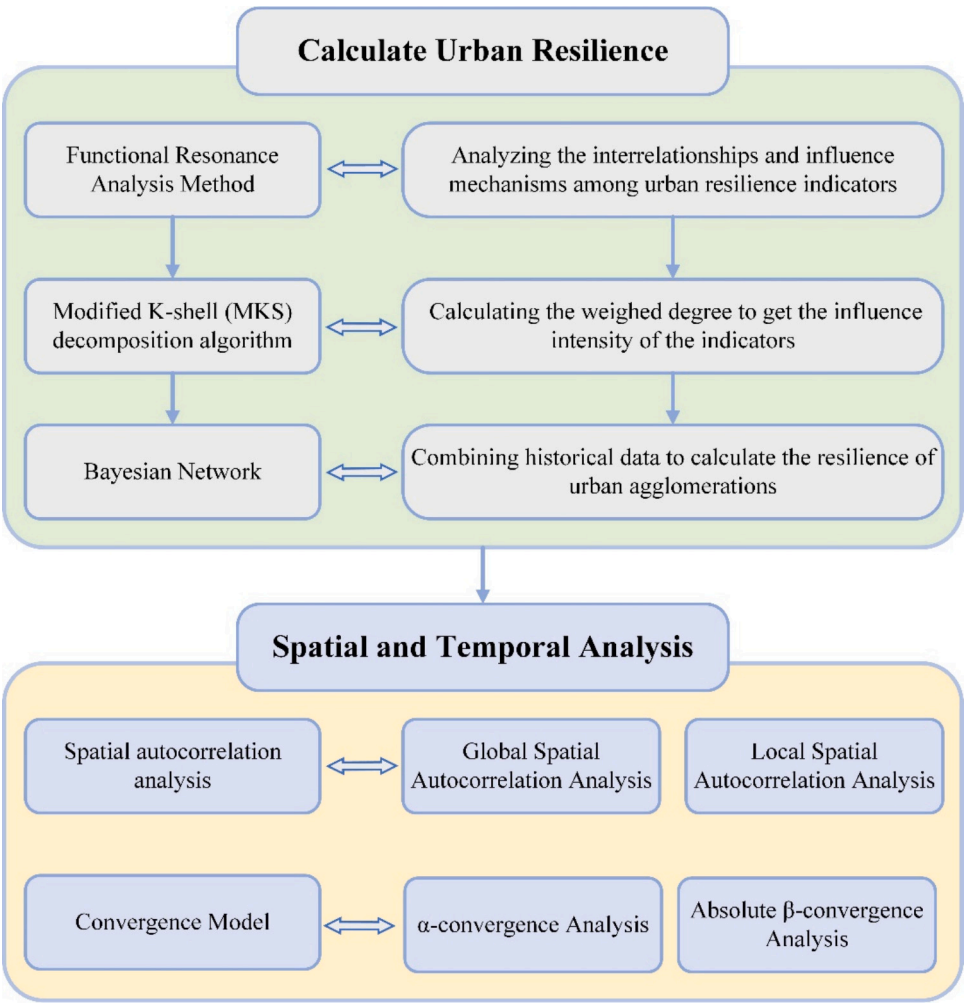


Fig. 3. Research framework.

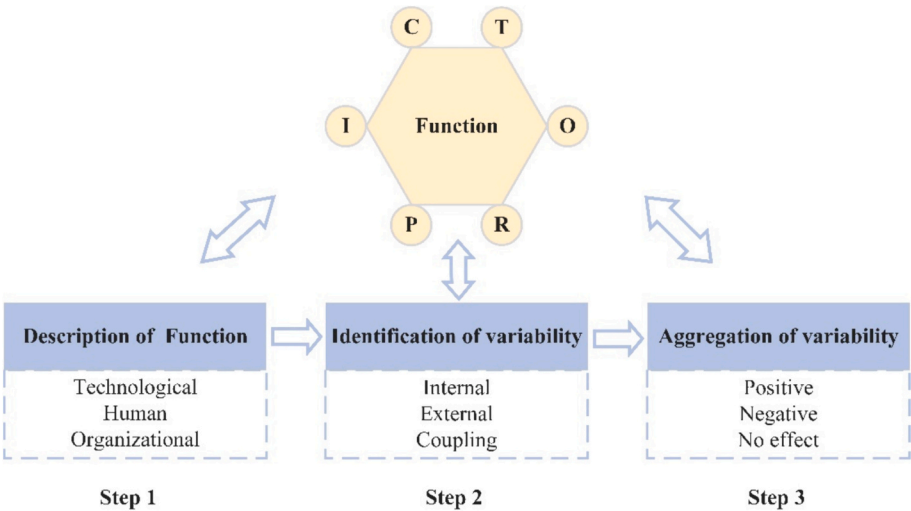


Fig. 4. General procedure for FRAM modeling.

is crucial to grasp the interplay of functions and the impact of variability from upstream functions on downstream ones. Special focus is given to the variability of function outputs, which can stem from three sources: internal (endogenous), external (exogenous), and coupling between upstream and downstream functions.

The concluding phase of FRAM modeling involves the aggregation of variability, a process that establishes interconnections among distinct functions, thereby creating couplings between them. As depicted in Fig. 5, coupling implies that the output of one function can influence various aspects of other functions (such as inputs, preconditions,

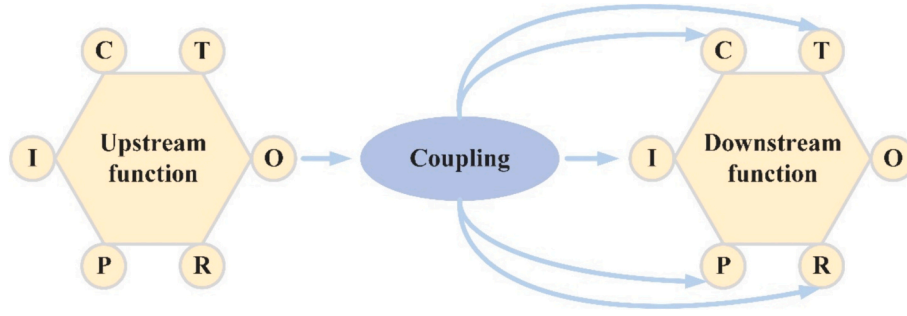


Fig. 5. Coupling process of variability.

resources, controls, or time). This stage highlights the manner in which variability in one function impacts other functions, and elucidates how the interaction of two or more functions can lead to instability and unpredictable outcomes.

4.2. Modified K-shell

In graph theory, a typical complex network is represented as defined in Eq. (1). The k-shell decomposition algorithm is a coarse-grained decomposition method that leverages the global network structure to assess the strength of relationships between nodes based on their positions within the network. The process of this method is illustrated in Fig. 6. In this study, we have introduced the concept of weighted degree, which fundamentally accounts for the strength of relationships between nodes and takes into consideration the varying importance of connected neighbors as measured by the potential influence of the edges. This enhancement addresses the limitation of equal contributions from neighbors, resulting in a more precise identification, as demonstrated in Eq. (4).

$$G = (V, E) \quad (1)$$

where $V = \{v_1, v_2, \dots, v_n\}$ and $E = \{e_1, e_2, \dots, e_n\}$ are the set of nodes and the set of edges in the complex network, respectively. Then Eq. (1) can be mathematically interpreted in terms of the adjacency matrix A .

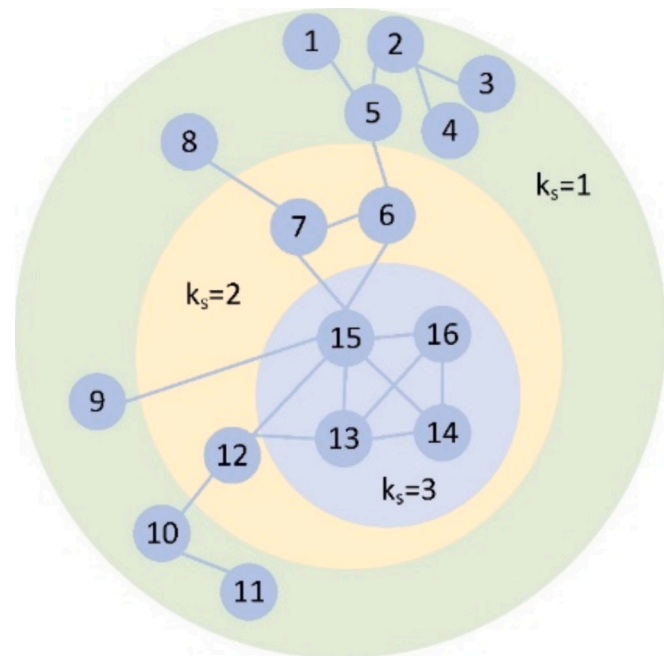


Fig. 6. K-shell decomposition example diagram.

$$A_{ij} = \begin{cases} a_{ij} \cdot w_{ij} & i \rightarrow j \\ 0 & \text{else} \end{cases} \quad (2)$$

where a_{ij} is equal to 1 when the output of node v_i is an aspect of node v_j and zero otherwise. w_{ij} is the weight of the edges connecting v_i and v_j , as shown in Eq. (3).

$$w_{ij} = K_i + K_j \quad (3)$$

where K_i and K_j represent the degree values of nodes i and j .

$$WK(i) = \sum_{j \in N_i} I_{ij} k_s(j) + \sum_{j \in N_i} w_{ij} \quad (4)$$

where I_{ij} represents the influence coefficient of edges between nodes i and j , as shown in Eq. (5). $k_s(j)$ represents the level in which j is after k-shell decomposition. N_i represents the set of all neighbors of node i . w_{ij} is the weight of edges between nodes i and j .

$$I_{ij} = \frac{|N_i \cap N_j| + 1}{|N_i \cup N_j| + 1} \quad (5)$$

where N_i and N_j represent the set of neighboring nodes of node i and node j , $N_i \cap N_j$ represents the same neighbors of nodes i and j , and $N_i \cup N_j$ represents the total neighbors of i and j and does not include nodes i and j .

4.3. Bayesian network

Although the Functional Resonance Analysis Method (FRAM) effectively reveals the interactions and dynamics among factors influencing urban resilience, it inherently remains a subjective qualitative analysis tool and cannot provide quantitative data. To overcome these limitations, we have introduced Bayesian Networks (BNs) as a complement to FRAM. BNs consist of Directed Acyclic Graphs (DAG) and Conditional Probability Tables (CPTs). They are particularly adept at merging various sources of information, including empirical data, expert judgment, and statistical models, enabling the dynamic update of probabilities and expected outcomes. This method not only enhances the evidence base of FRAM analysis but also eliminates the subjective biases introduced by experts, providing a quantitative tool for assessing urban resilience. A key advantage of BNs over traditional regression algorithms is their capability for bidirectional inference, exemplified by Eq. (6) for forward reasoning and Eq. (7) for backward reasoning.

$$P(T = t) = P(T = t | X_1 = x_1, \dots, X_n = x_n) \times P(X_1 = x_1, \dots, X_n = x_n) \quad (6)$$

$$P(X_i | T = t) = \frac{P(X_i = x_i) P(T = t | X_i = x_i)}{P(T = t)} \quad (7)$$

where $t = t_1, t_2, \dots, t_p$ is a range of P states for the node T ; $x = x_1, x_2, \dots, x_n$ is the variables; $P(T = t | X_1 = x_1, \dots, X_n = x_n)$ denotes the conditional probability distribution of T ; $P(X_1 = x_1, \dots, X_n = x_n)$ represents the joint

probability distribution of X_i .

4.4. Spatial autocorrelation analysis

The concept of spatial autocorrelation is critical in understanding how geographic phenomena are correlated with one another across space. The Moran's I test, established by Moran (1950), serves as a pivotal measure for assessing spatial correlation. This metric is instrumental in identifying patterns of spatial autocorrelation, whether they manifest as clustered, dispersed, or random spatial arrangements. The rationale for employing the Moran's I test in this study stems from its ability to quantify the degree of spatial correlation, offering insights into the spatial spillover effects among urban agglomerations. By applying the global Moran's I index, our analysis seeks to elucidate the spatial correlation and disparities in resilience levels across urban agglomerations, further substantiated by a significance test. This approach enables a nuanced understanding of how resilience attributes are distributed spatially, which is crucial for devising targeted interventions.

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n W_{ij}} \frac{\sum_{i=1}^n \sum_{j=1}^n W_{ij} (R_i - R)(R_j - R)}{\sum_{i=1}^n (R_i - R)^2} \quad (8)$$

where R_i represents the resilience value of the i th city. R represents the average resilience value of the 16 cities. n is the number of cities in the urban agglomeration. W is an $n \times n$ spatial weight matrix, where W_{ij} represents the elements of the matrix. In this study, we utilize a widely-used rook contiguity spatial weights matrix, whose elements equal 1 if two cities share common borders, and 0 otherwise (He & Jiang, 2021).

4.5. Convergence models

4.5.1. α -convergence

α -convergence explores the dynamics of regional variability over time, serving as a gauge for the homogenization of economic or other measurable indicators across different regions. A diminishing trend in variability, or α -convergence, suggests that regions are becoming more similar over time, indicative of a possible equalization in the measured variable. The rationale behind selecting α -convergence for this study lies in its ability to provide a straightforward statistical measure of such equalization trends, employing the coefficient of variation as detailed in Eq. (9). This method offers a clear, quantifiable means to assess whether the disparities in urban resilience are narrowing, a vital step in understanding the broader patterns of urban development and policy impact.

$$\alpha = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (R_i - R)^2}}{R} \quad (9)$$

4.5.2. Absolute β -convergence

The concept of β -convergence, originating from neoclassical economic growth theory (Solow, 1956), posits that regions with lower initial metrics (e.g., per capita income, resilience levels) tend to grow faster than their more affluent counterparts, leading to a convergence over time. Absolute β -convergence is examined to determine if an inverse relationship exists between the initial urban resilience level and its subsequent growth rate. This study advances the traditional analysis of β -convergence by incorporating spatial considerations, acknowledging that spatial correlation features prominently in urban resilience dynamics. By integrating spatial factors and control variables into the β -convergence model, we evaluate the need for either a Spatial Error Model (SEM) or a Spatial Lag Model (SLM) through empirical validation, as represented in Eqs. (10) to (12). This enhancement is crucial for capturing the spatial dependencies that might influence convergence processes, thus providing a more accurate and comprehensive

understanding of urban resilience evolution.

$$\ln(R_{i,t+1}/R_{i,t}) = \alpha + \beta \ln R_{i,t} + \varepsilon_{i,t} \quad (10)$$

$$\ln(R_{i,t+1}/R_{i,t}) = \alpha + \beta \ln R_{i,t} + \rho W(\ln(R_{i,t+1}/R_{i,t})) + \varepsilon_{i,t} \quad (11)$$

$$\ln(R_{i,t+1}/R_{i,t}) = \alpha + \beta \ln R_{i,t} + (I - \lambda W)^{-1} \mu_{i,t} \quad (12)$$

Eq. (10) represents the β -convergence model, where $R_{i,t}$ denotes the resilience level of region i in time period t , α is a constant term, β represents the coefficient of convergence, and $\varepsilon_{i,t}$ represents the random error. Eq. (11) is the SLM, and Eq. (12) is the SEM. In these models, W is the spatial weight matrix, ρ is the spatial lag coefficient, indicating the spatial dependence of the observed target. λ represents the spatial error coefficient, and $\mu_{i,t}$ is a random error term normally distributed.

5. Case study

5.1. Study area and data sources

The Chengdu-Chongqing urban agglomeration, positioned in the upper reaches of the Yangtze River, spans across Sichuan Province and Chongqing Municipality. This region encompasses the principal cities of both Sichuan Province and Chongqing Municipality, including Chengdu, Deyang, Mianyang, Meishan, Ziyang, Suining, Leshan, Ya'an, Zigong, Luzhou, Neijiang, Nanchong, Yibin, Dazhou, Guang'an, and Chongqing, as depicted in Fig. 7. Representing about 7 % of China's national population and economic output, this agglomeration stands as the foremost developed and promising urbanization zone in Western China. It plays a vital role as a connecting hub for the Yangtze River Economic Belt and the 'Belt and Road' initiative. However, its basin location makes it vulnerable to a cascade of natural disasters like earthquakes, landslides, and floods, posing risks to the supply chain and infrastructure in cities, including those outside the immediate hazard zones (Lu, Zhang, et al., 2022). The data for this study were sourced from the 'Sichuan Statistical Yearbook (2012–2021)' and the 'Chongqing Statistical Yearbook (2012–2021)'. Focusing on Chengdu as a case study, the dataset is illustrated in Table 2, with the comprehensive data for all 16 cities provided in Supplementary Material.

5.2. Computation of urban agglomeration resilience

5.2.1. Development of FRAM model

In this study, the FRAM introduced in Section 4.1 was used to analyze the measurement indicators collected in Section 3. First, six experts in the field of urban resilience were invited to participate in two rounds of expert discussions. In the first round, each indicator was described according to the reference literature. In the second round of expert discussions, the focus was on identifying the interconnections between each indicator and their inherent variability. Based on the principles of identifying functional variability discussed in Section 4.1, this study particularly emphasized the coupling of upstream and downstream functions in this case study. This coupling was determined through a comprehensive analysis of all six aspects of the listed indicators, as summarized in Table 3. The couplings identified in Table 3 were then integrated into FRAM model visualization (FMV), developing the instantiated model used in this study, as illustrated in Fig. 8.

5.2.2. Topology analysis of complex network

Complex networks can be developed based on the FRAM model, with functionally equivalent nodes and directed connections representing couplings. This section constructs a complex network from the FRAM model shown in Fig. 8, aimed at quantitatively analyzing the selected resilience indicators to determine the interrelationships and importance of urban resilience factors across different aspects. First, the adjacency matrix of the complex network based on the FRAM model is obtained as

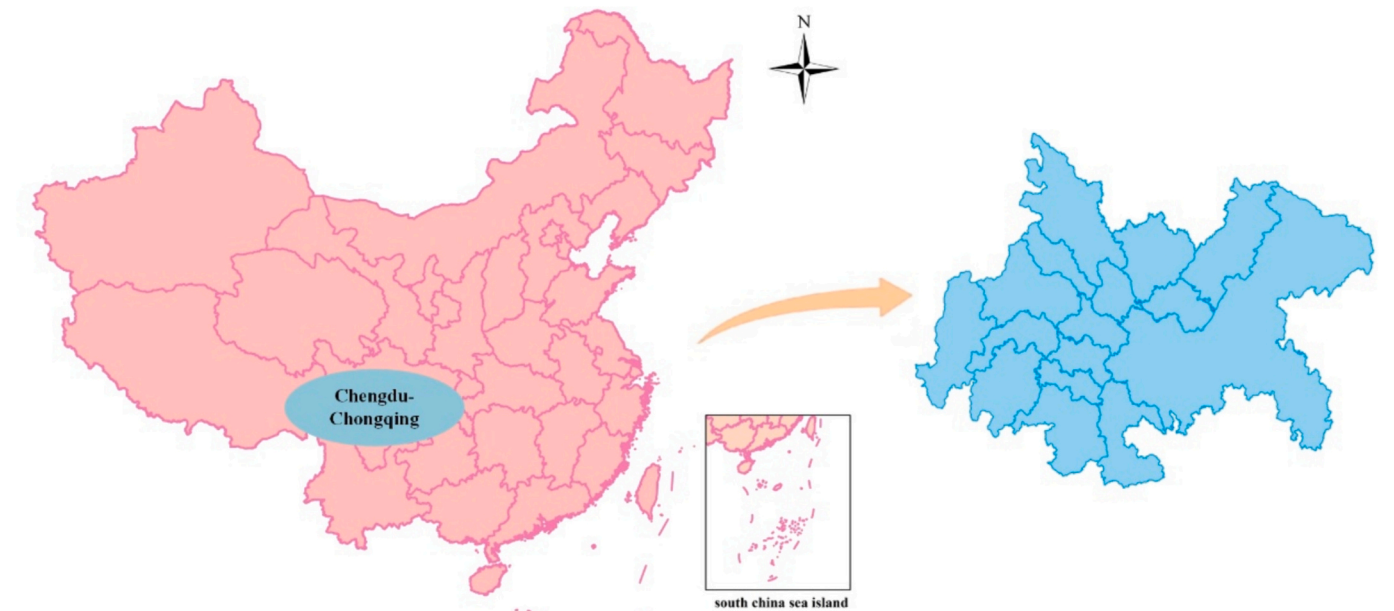


Fig. 7. Study area.

Table 2
The original data of Chengdu.

Evaluation Index	2021	2020	2019	2018	2017	2016	2015	2014	2013	2012
F1	217.3	213.8	189.2	176.5	159.2	174.8	105.2	83.9	73.7	58.2
F2	98.15	93.22	98.17	84.03	81.74	79.16	75.58	72.93	70.17	68.63
F3	1478.3	1461	1157	1139	1119	1110	1209.5	1190.5	1180	1181.5
F4	79.5	77	74.4	73.12	71.9	70.6	71.47	70.3	69.4	68.44
F5	3.6	1.7	3.6	3.2	0.7	5	5.4	4.65	2.71	0.12
F6	9.46	8.63	8.46	8.1	7.38	6.51	6.06	6.11	5.75	5.42
F7	66.4	65.7	65.6	51.4	53.2	53.1	52.8	51	50.2	49.1
F8	5.2633	4.8593	4.5878	4.2128	3.8918	3.5902	3.3476	3.2665	2.9968	2.7194
F9	1697.9	1520.4	1483	1424.2	1275.5	1175.4	1157.6	1025.2	898.5	781
F10	26.4	25.5	26.4	27.6	25.3	27.9	26.4	24.3	25.8	21.9
F11	462.48	416.8	366.16	325.5	277.4	190.8	156.2	220.3	187	171.8
F12	16.6	18.7	15.87	14.2	14.06	13.89	14.62	14.78	15.85	16.24
F13	206.01	206.68	197.13	192.09	182.13	161.24	159.14	157.87	156.46	153.88
F14	12,497	11,954	12,121	10,755	10,183	9853	8481	8190	7976	7605
F15	10.65	9.48	9.31	8.8	7.89	8.2	7.9	8.08	7.79	7.46
F16	245.29	280.44	306.65	236.08	299.97	351.2	338.34	380.84	322.56	317.58
F17	44.3	43.75	43.5	41.3	41.6	39.9	38.7	35.9	35.7	35.1
F18	39.8	41	43	51	56	63	64.05	77	78.3	79.2
F19	81.9	76.5	78.6	70.3	64.9	63.5	62.6	61.1	60.8	78.1
F20	97.4	95.4	90.7	85.5	83.2	70.5	79.5	68.1	65.4	71.1
F21	1.32	1.21	1.34	1.54	1.26	1.31	0.94	0.76	0.65	0.58
F22	770	720	686	681	667	650	644	635	631	629
F23	16.1	16.1	15.9	15.8	14.57	13.25	11.88	10.34	8.76	8.03
F24	0.337	0.324	0.322	0.32	0.289	0.286	0.285	0.277	0.277	0.277
F25	11.2	10.4	9.5	8.7	7.5	6.4	6.3	6.2	5.1	4.6

shown in Appendix A. Then, using the node information in the adjacency matrix, the degree, k_s value, and weighted degree of each node in the complex network are calculated sequentially using the MKS method outlined in Section 4.2, as illustrated in Fig. 9 and Fig. 10.

Fig. 9 (The phase diagrams shown in Fig. 9(b) are projections of Fig. 9(a) along the Y-Z axis.) reveals that most of the nodes have out-degree or in-degree equal to 3, and the number of nodes with the range of the total degree falling within the interval [4,6] accounts for 52 % of the total number of nodes, which indicates that the network has a high degree of activity, and that there exists a close connection between the indicators affecting the urban resilience, which has an important impact on the final resilience results. In addition, as shown in Fig. 10(a), nodes with k_s of 4 account for 68 % of all nodes. This pattern indicates a relatively even distribution of the influence nodes and a reasonable

construction of the urban resilience influence indicator system. However, only the macro-analysis of degree and k_s value cannot fully distinguish the influence of each node in the network, Fig. 10(b) weighted degree shows the importance of each resilience indicator at the micro level, which effectively determines the size of the influence of each indicator node in the complex network, and creates the conditions for the establishment of the conditional distribution of the BN model.

5.2.3. Establishment of Bayesian Networks

Building on the urban resilience assessment framework and indicator system introduced in Section 3, this study utilized the Bayesian modeling software AgenaRisk for visualizing the model results, which are displayed in Fig. 11. Subsequently, combining the analysis from FRAM with the outcomes derived from MKS, the urban data were input

Table 3

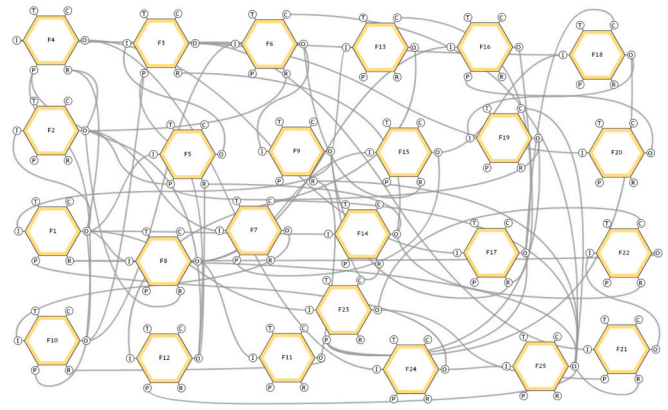
Identification of functions and coupling variability in urban resilience indicators.

Function	Description	The coupling of variability
F1	Measuring the degree of perfection of social security	F9(O), F22(O), F24(O), F25(O)→F1(O)→F2(I), F4(R), F5(I), F8(I), F10(P), F14(I), F23(I)
F2	Reflecting the potential of the social system	F1(O)→F2(O)→F4(R), F5(P), F6(R), F7(I), F8(R), F5(O), F10(O), F12(O), F14(O)→F3(O)→F18(I), F20(I), F21(I)
F3	Reflecting a city's disaster resilience	F1(O), F2(O), F6(O), F10(O), F12(O)→F4(O)→F17(I), F24(I)
F4	Characterizing a city's disaster response capability	F1(O), F2(O), F12(O), F14(O), F19(O)→F5(O)→F3(I)
F5	Reflecting the city's population carrying capacity	F2(O), F10(O), F25(O)→F6(O)→F4(I), F9(I), F15(R)
F6	Reflecting the development level of the urban economy	F2(O), F15(O), F25(O)→F7(O)→F10(I)
F7	Reflecting the rationality of the urban economic structure	F1(O), F2(O)→F8(O)→F11(I), F15(I), F16(I)
F8	Revealing the economic vitality of the city and the quality of life of the population	F6(O), F11(O), F25(O)→F9(O)→F1(I), F13(I), F14(R), F17(R)
F9	Representing the payment capability for urban emergency rescue	F1(O), F7(O), F11(O)→F10(O)→F3(P), F4(P), F6(I)
F10	Reflecting the stability and growth potential of the urban economy	F8(O)→F11(O)→F9(R), F10(R)
F11	Measuring the external economic dependency of the city	F13(O), F25(O)→F12(O)→F3(P), F4(P), F5(R)
F12	Reflecting the rationality of urban road resource allocation	F9(O), F17(O)→F13(O)→F12(I)
F13	Reflecting the response capacity of urban infrastructure after disasters	F1(O), F9(O)→F14(O)→F3(R), F5(R), F22(R)
F14	Measuring the urban emergency rescue and medical response capability	F6(O), F8(O)→F15(O)→F7(P), F18(I)
F15	Reflecting the completeness of infrastructure and power supply stability	F8(O), F20(O)→F16(O)→F17(R)
F16	Representing water supply stability and the ability to prevent post-disaster disease spread	F4(O), F9(O), F16(O)→F17(O)→F13(C), F18(C), F23(P)
F17	Describing the ecological self-purification capacity of the city	F3(O), F15(O), F17(O)→F18(O)→F19(I), F23(P)
F18	Reflecting the degree of air pollution in the city	F18(O), F25(O)→F19(O)→F5(P), F23(P)
F19	Representing the overall air quality of the city	F3(O)→F20(O)→F16(P)
F20	Measuring the quality and availability of urban water resources	F3(O), F23(O), F25(O)→F21(O)→F22(I)
F21	Reflecting the institutional preparedness for disaster prevention	F14(O), F21(O), F23(O)→F22(O)→F1(R)
F22	Describing the distribution and accessibility of municipal resources	F1(O), F17(O), F18(O), F19(O)→F23(O)→F21(P), F22(C)
F23	Representing the effectiveness of municipal services and public satisfaction	F4(O)→F24(O)→F1(P), F25(I)
F24	Reflecting the coordination capability of the organizational system	F24(O)→F25(O)→F1(R), F6(C), F7(C), F9(R), F12(P), F19(C), F21(R)
F25	Reflecting the degree of participation of organizational management personnel	

into the BN to obtain quantified resilience results. The specific steps are as follows:

Step 1 Determination of the prior probability distribution of the root node.

The historical data collected in Section 5.1 had to be transformed into a priori probability values in order to be input into the BN model. For dimensionless indicators, such as F4 (Urbanization Rate of the Population), we control the data between [0.1] through normalization and enter them directly into the root node as a percentage. For contin-

**Fig. 8.** FRAM model of urban resilience indicators.

uous indicators with scales, such as F6 (GDP Per Capita), the Truncated Normal Distribution (TNORM) method as shown in Fig. 12, was used to transform the data set into a prior probability distribution for the node. Using the data in Table 2 and the node F6 as an example, the following Eq. (13) was used to model the prior probability distribution of the indicator, and the results are shown in Fig. 13.

$$F6 : (fit \sim TNORM) \sim (\mu = a, \sigma^2 = b, UB = \max(F6) \text{ } LB = \min(F6)) \quad (13)$$

where μ represents the mean value of F6; σ^2 is the variance of F6; LB denotes the minimum value of F6; UB represents the maximum value of F6.

Step 2 Determination of Conditional Probability Table.

Based on the prior probability distribution of the root node obtained in the first step, the conditional probability tables for the intermediate nodes are calculated using the weighted degree determined by MKS (as shown in Fig. 10 (b) and Eq. (14)) and predefined Boolean expressions (Hosseini & Barker, 2016). The results are listed in Table 4.

$$\chi(j) = \sum_{i \in j} \frac{WK(i)}{\sum WK(i)} \Lambda(i) \quad (14)$$

where $\Lambda(i)$ denotes the prior probability of the i th root node pointing to the j th intermediate node.

Step 3 Calculation results of BN network model.

In the final step, annual historical statistics for 16 cities are converted into prior probabilities for the root nodes. These probabilities, combined with the conditional probability distribution expression from Table 4, are then input into the developed BN model to calculate the resilience values of the cities. As an illustration, using the 2021 data for Chengdu, the model computes the city's resilience value as 88.768 %, with the results being displayed in Fig. 14. It should be noted that the absolute resilience values derived from BN simulations are not heavily significant in a physical sense. In the context of this study, a higher resilience value for a city is indicative of its resistance to disturbance.

5.3. Results

5.3.1. The results of urban agglomeration resilience

As detailed in Section 5.2, the resilience values for each city in the Chengdu-Chongqing urban agglomeration from 2012 to 2021 were calculated and are presented in Table 5. The data indicate a notable and consistent upward trajectory in the resilience values of all cities over the last decade. This trend has been especially pronounced since 2016, coinciding with the State Council's approval of the 'Chengdu-Chongqing Urban Agglomeration Development Plan' in March 2016, aimed at fostering new urbanization in the western region. The subsequent 'Reply to the Chengdu-Chongqing Urban Agglomeration Development Plan' by

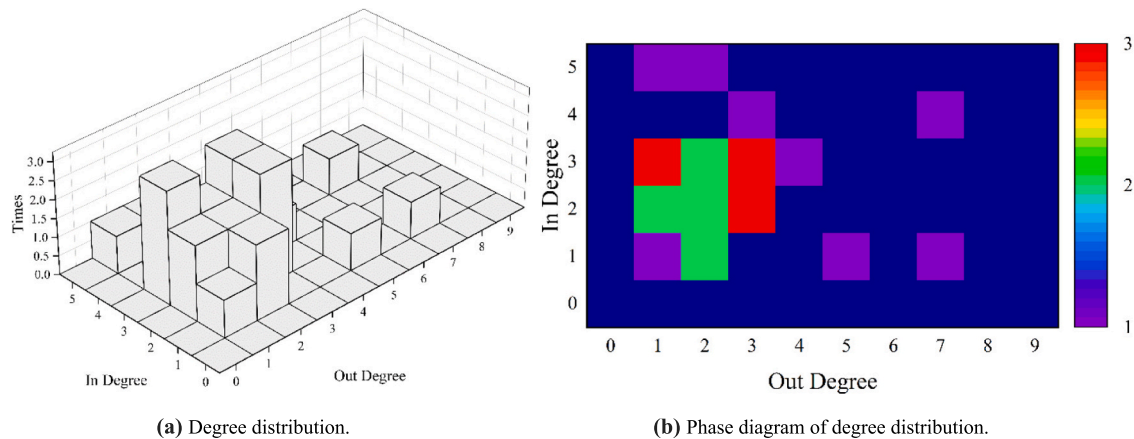


Fig. 9. Degree distribution of the developed CN.

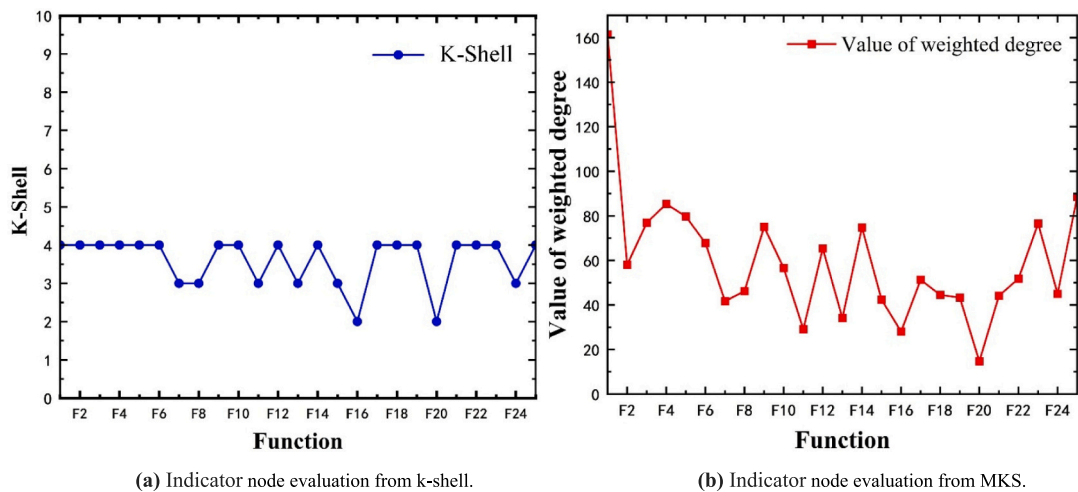


Fig. 10. Indicator node evaluation from different perspectives.

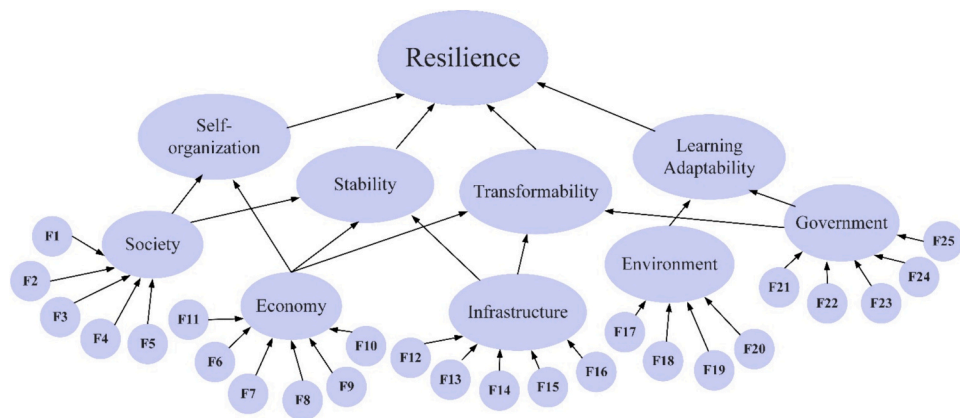


Fig. 11. BN developed for urban resilience evaluation.

the State Council in April 2016 further contributed to the enhancement of urban resilience.

The growth patterns observed across the cities are markedly diverse, reflecting the unique economic, social, institutional, and infrastructural contexts of each city. While the general trend has been upward for most cities, the rates of growth differ considerably. Cities like Luzhou and Yibin have experienced more stable and gradual increases in resilience.

In contrast, cities such as Neijiang and Suining have shown significant spikes in resilience in certain years, likely attributable to targeted initiatives or investments.

5.3.2. α -convergence analysis

In this section, we focus on analyzing the spatial distribution changes over time of cities in the Chengdu-Chongqing urban agglomeration.

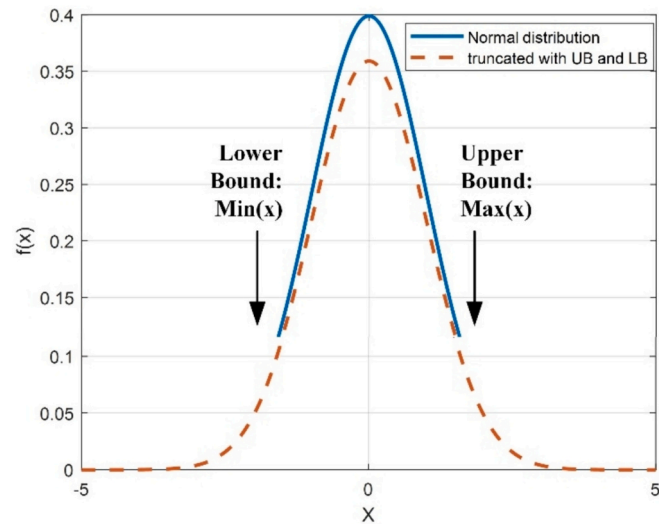


Fig. 12. Illustrative example of a truncated normal distribution.

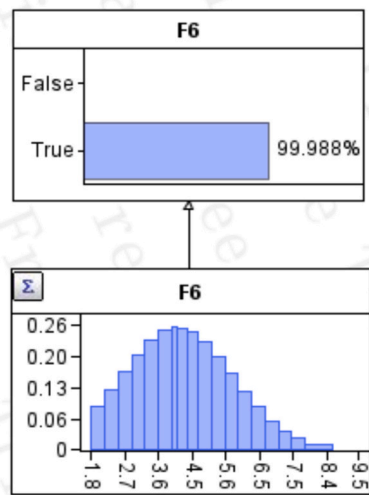


Fig. 13. Transformation results of node F6.

Fig. 15 reveals the spatial distribution and its evolution of resilience in the Chengdu-Chongqing urban agglomeration for the years 2012, 2017, and 2021. In 2012, Chengdu and Chongqing, as the core cities of the agglomeration, had significantly higher resilience levels than other cities, presenting a distinct resilience fault line on the map. This phenomenon reflected the uneven regional development at that time and the impact of resource allocation and policy orientation on urban resilience. By 2017, other cities, including Nanchong and Mianyang, began to show significant improvement in resilience, gradually closing the gap with Chengdu and Chongqing, indicating a trend towards coordinated regional development. By 2021, this trend was further consolidated and strengthened, with the map clearly showing a more uniform level of resilience across the entire urban agglomeration, indicating a more balanced development trajectory.

To scientifically validate our observations, we calculated α values for 16 cities within the Chengdu-Chongqing urban agglomeration, spanning the years 2012 to 2021, using Eq. (9). The results, as depicted in Fig. 16, demonstrates a considerable decline in α values over this period, suggesting a movement towards uniformity among these cities. This pattern affirms our preliminary finding that the gap in resilience within the cities of the Chengdu-Chongqing cluster is narrowing. Moreover, our

findings are consistent with the research conducted by Lu, Zhang, et al. (2022), thus enhancing the credibility of our study in the context of the existing scholarly work in this area. The observed diminution in these disparities is likely attributable to the reevaluation of urban vulnerability in the Sichuan Basin, initiated by the Wenchuan earthquake in 2012. This reevaluation has led to the implementation of integrated, long-term resilience strategies across the region.

5.3.3. Spatial autocorrelation analysis

(1) Global spatial autocorrelation analysis.

Time-based analysis alone is insufficient to gauge the extent of clustering and mutual influence among urban resilience levels across spatial dimensions. Hence, this section focuses on uncovering the spatial aggregation and interrelationships of resilience levels among cities within the Chengdu-Chongqing urban agglomeration. To this end, the global Moran's I index, a statistical tool for measuring spatial data autocorrelation, is employed. This index evaluates the similarity between the resilience values of a region and its neighboring areas within a specific spatial distribution. The Moran's I index ranges from $[-1, 1]$, where a value greater than 0 implies positive spatial correlation, less than 0 indicates negative spatial correlation, and a value of 0 suggests no spatial correlation.

For the period from 2012 to 2021, the global Moran's I index for the Chengdu-Chongqing urban agglomeration is calculated and presented in the Table 6. The findings reveal: (1) Throughout the decade, the Moran's I index has consistently been negative, indicating a negative spatial correlation in urban resilience within the agglomeration. This suggests that high resilience in one city may adversely affect the resilience development of its neighboring cities. (2) There has been a year-on-year decrease in the Moran's I index. While the index remained negative from 2012 to 2016, its p -value was above 1 %, making it statistically insignificant. However, from 2017 onwards, not only has the negative value of the Moran's I index increased, but its p -value has also dropped significantly below 1 %. This change underscores the growing statistical significance of the negative spatial correlation conclusion.

(2) Local spatial autocorrelation analysis.

The global Moran's I index analysis indicates a negative spatial correlation within the Chengdu-Chongqing urban agglomeration. However, this index alone does not fully capture the spatial clustering characteristics among the different cities. To delve deeper into this aspect, this section employs Moran's I scatter plots to examine the local spatial clustering of urban resilience within the agglomeration.

From the Moran's I scatter plot analyses for 2017 and 2021, illustrated in Fig. 17, two notable local spatial clustering patterns emerge within the Chengdu-Chongqing urban agglomeration: low-high clustering in the second quadrant and high-low clustering in the fourth quadrant. Cities in the low-high pattern, while having lower resilience indices, are surrounded by cities with higher resilience indices. In contrast, cities in the high-low pattern exhibit higher resilience but are encircled by cities with lower resilience indices. These variations in local spatial clustering reflect the intricate spatial dependencies formed by cities influencing each other's resilience levels and development statuses.

Specifically, the 2017 data in Fig. 17(a) show cities like Ya'an, Ziyang, Luzhou, and Dazhou in the low-high clustering second quadrant,

Table 4
Probability distribution expression.

Node	Probability distribution expression
Society	$0.3498F1 + 0.1258F2 + 0.1667F3 + 0.1849F4 + 0.1727F5$
Economy	$0.2141F6 + 0.1317F7 + 0.1461F8 + 0.2371F9 + 0.1788F10 + 0.0920F11$
Infrastructure	$0.2672F12 + 0.1394F13 + 0.3055F14 + 0.1731F15 + 1146F16$
Environment	$0.3334F17 + 0.2895F18 + 0.2815F19 + 0.0954F20$
Government	$0.1441F21 + 0.1693F22 + 0.2502F23 + 0.1470F24 + 0.2892F25$

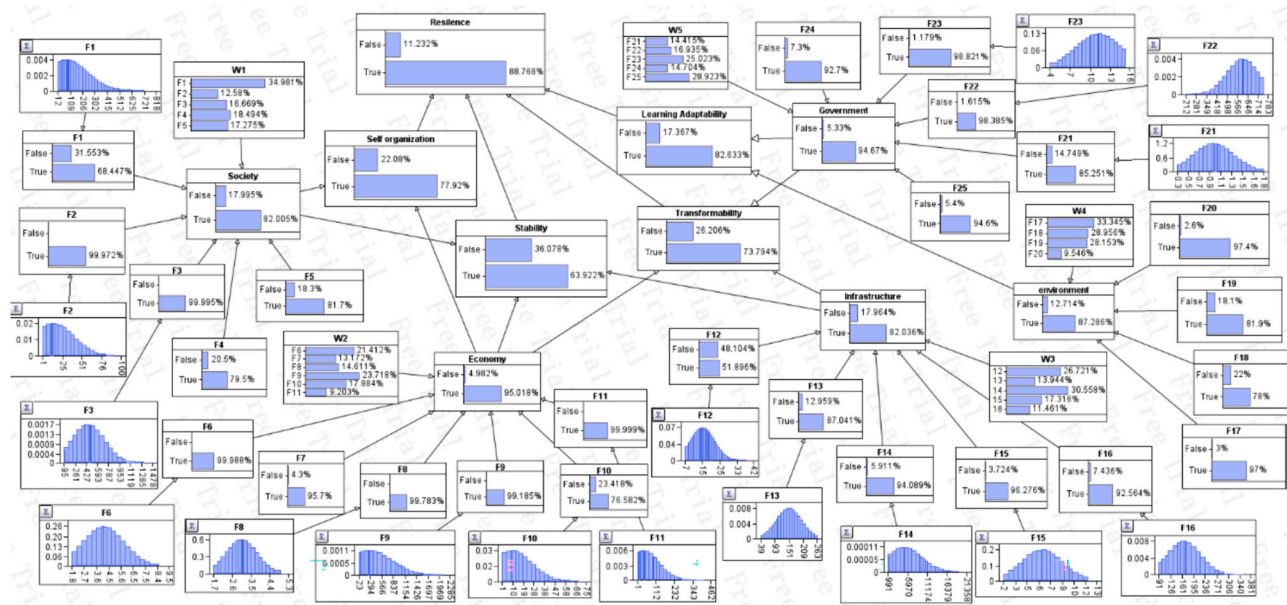


Fig. 14. BN model of Chengdu in 2021.

with Chengdu and Chongqing positioned in the high-low clustering fourth quadrant. By 2021, as depicted in Fig. 17(b), the low-high clustering phenomenon expanded, encompassing more cities including Ya'an, Ziyang, Luzhou, Dazhou, Deyang, and Guang'an.

An integrated analysis with Fig. 15 reveals that cities in the low-high clustering pattern are directly connected to Chengdu and Chongqing, the central hubs of the Chengdu-Chongqing urban agglomeration. These core cities, with their higher development levels and resilience, can significantly influence the surrounding lower-resilience cities. This influence may manifest through direct or indirect economic and social mechanisms. For instance, the rapid advancement of core cities could lead to a concentration of resources and talent, thereby posing challenges for neighboring cities, such as unequal resource distribution and diminished development opportunities. This scenario might also highlight disparities in regional development strategies among the cities in the agglomeration.

5.3.4. Absolute β -convergence analysis

In this section, the absolute β -convergence model is used to analyze the spatial convergence of the resilience level of Chengdu-Chongqing urban agglomeration. According to Moran's I test results in Section 5.3.3, it is found that the geographical distribution of urban resilience is

not random. Therefore, we consider the spatial spillover effect of urban agglomeration resilience in the convergence model. The test results for absolute β -convergence are calculated by Eq. (10), Eq. (11) and Eq. (12), as shown in Table 7.

Table 7 illustrates the results of the absolute β -convergence test spanning 2012 to 2020. It reveals that neither the LM(LAG) nor LM (ERR) statistics achieved statistical significance, thereby hindering a direct determination of the most suitable spatial model. Consequently, a robustness analysis was undertaken. The results of this analysis indicated that the R-LM(ERR) statistic was lower than R-LM(LAG), suggesting that the Spatial Lag Model (SLM) is more appropriate for spatial convergence analysis in comparison to the Spatial Error Model (SEM). Further, a comparative evaluation of log-likelihood test values (Log(L)) and goodness of fit (R^2) revealed that the SLM's performance from 2012 to 2021 surpassed that of the traditional absolute β -convergence model, demonstrating its superior fit for the data under consideration.

Analysis of the SLM results reveals a β value less than 0, signifying a clear absolute β -convergence trend in the resilience of the Chengdu-Chongqing urban agglomeration, with spatial effects considered. This supports conclusion of Wang and Yan (2023) that cities in the Chengdu-Chongqing urban agglomeration, initially with lower resilience levels, are currently experiencing a relatively rapid growth rate. As a result, the

Table 5
The urban resilience of Chengdu-Chongqing urban agglomeration from 2012 to 2021.

City	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Chendu	29.231	36.515	43.08	49.593	59.522	64.234	72.688	82.751	86.136	88.768
Dazhou	0.35	0.48	1.028	2.127	3.228	4.907	7.036	9.058	12.117	15.417
Deyang	1.428	2.142	2.479	4.144	5.553	8.751	12.778	18.142	21.675	22.372
Guang'an	0.48	1.066	2.515	3.144	4.909	7.287	11.211	16.015	18.757	20.947
Leshan	0.77	1.477	2.202	2.881	4.723	7.781	11.944	16.168	19.758	24.472
Luzhou	0.68	1.229	2.007	2.621	3.579	5.852	8.597	11.267	21.108	19.997
Meishan	0.86	1.475	2.209	3.824	5.212	9.019	12.237	17.34	19.174	23.826
Mianyang	1.041	1.426	2.268	4.248	6.981	14.065	15.168	21.926	25.54	29.96
Nanchong	1.164	2.043	2.974	4.205	6.095	11.601	13.908	17.797	20.944	25.204
Neijiang	0.43	0.72	1.162	1.767	3.112	5.353	9.45	13.533	16.455	22.349
Suining	0.57	1.462	2.383	4.163	5.882	7.723	12.169	15.015	20.461	26.525
Ya'an	0.91	0.76	1.491	1.871	3.583	5.237	8.441	17.297	19.437	22.273
Yibin	0.69	1.211	2.354	2.975	3.947	6.803	11.032	15.177	18.236	22.359
Chongqing	33.12	37.49	44.471	52.088	59.831	66.442	75.071	82.611	85.869	87.929
Ziyang	0.32	0.63	1.065	1.622	2.562	3.581	3.546	6.46	8.399	10.921
Zigong	1.066	1.569	2.536	4.176	6.035	10.328	13.538	17.896	21.157	24.449

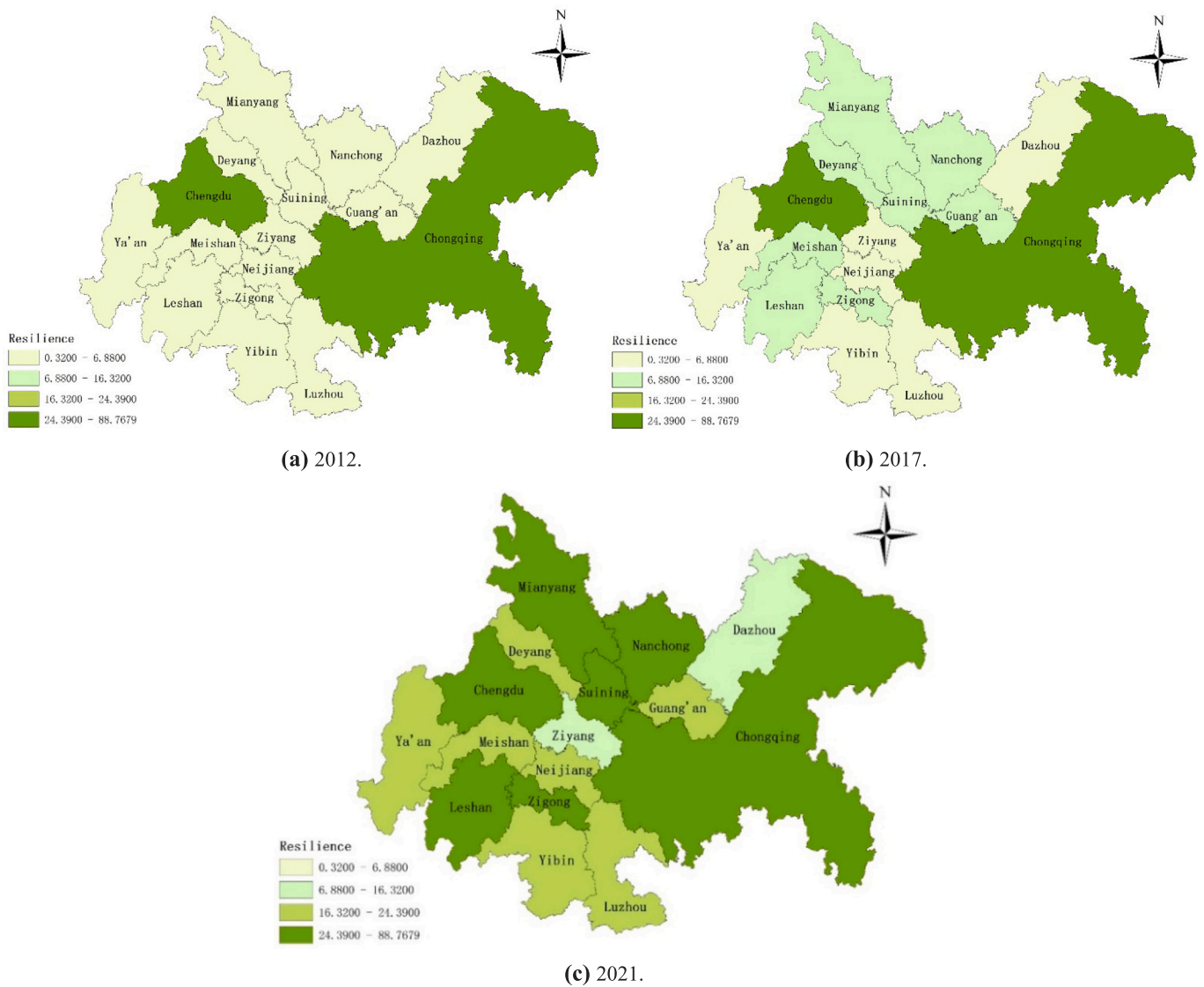


Fig. 15. Resilient spatial distribution maps of Chengdu-Chongqing urban agglomeration in 2012, 2017, and 2021.

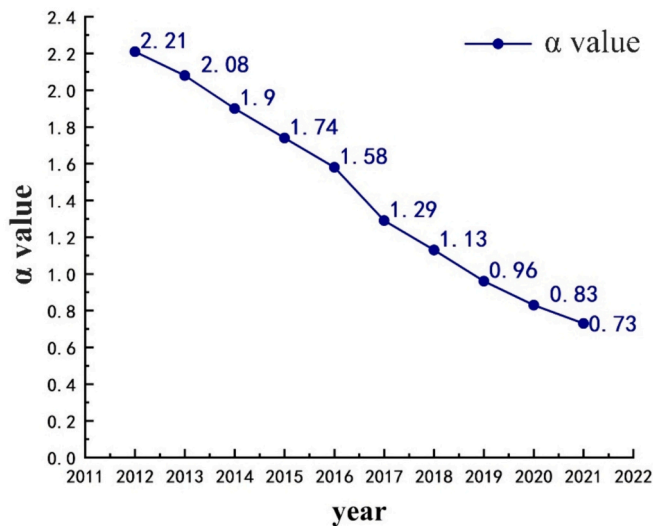


Fig. 16. Trend chart of α value for resilience in Chengdu-Chongqing urban agglomeration from 2012 to 2021.

resilience gap among these cities is reducing, evolving towards a more stable resilience equilibrium. From the Wenchuan earthquake in 2012, the Jiuzhaigou earthquake in 2017 and the Yibin earthquake in 2018, to the Qingyi River flood in 2020, cities have been learning and adjusting strategic responses to reduce the risk of natural disasters. The observed convergence trend points to an increasing harmonization in resilience capabilities across the agglomeration, marking a collective progression towards a more unified and resilient urban network.

5.4. Discussion

This study introduces a data-driven method to quantify urban resilience, utilizing the Chengdu-Chongqing urban agglomeration as a case study. Traditional models, as noted by Lu, Lu, et al. (2022), predominantly focus on qualitative evaluations and are often subject to subjective biases due to reliance on expert opinions. To mitigate this, our approach combines the FRAM, MKS decomposition algorithm, and BN algorithm, creating a unique blend of qualitative and quantitative analyses that leverages statistical data to significantly reduce the biases inherent in traditional expert assessments. Consequently, it enhances the robustness of urban resilience analysis and addresses the limitations of previous quantitative methods, such as expert scoring (Qasim et al., 2016), the analytic hierarchy process (Zhang et al., 2020), and the

Table 6

Global Moran's I and testing of resilience in Chengdu-Chongqing urban agglomeration from 2012 to 2021.

Years	Moran's I	p	z
2012	-0.1737	0.06	-0.9386
2013	-0.1785	0.044	-0.9047
2014	-0.1825	0.034	-0.9396
2015	-0.1880	0.028	-0.9824
2016	-0.1923	0.022	-1.012
2017	-0.247	0.004	-1.268
2018	-0.2583	0.004	-1.3182
2019	-0.2746	0.004	-1.4469
2020	-0.2949	0.003	-1.5513
2021	-0.340	0.003	-1.7705

entropy method (Zhao, Fang, et al., 2021), as noted in our literature review.

Recognizing that most current evaluation methods in the field pay insufficient attention to the multi-scale evaluation and mechanisms of urban resilience, often overlooking the interactions between its elements (Liu, Ma, & Wang, 2022), our study adopts the principles of Complex Adaptive Systems (CAS). We have selected indicators from five key dimensions—economic, social, environmental, government, and infrastructure—to ensure a multi-scale evaluation. This approach is in line with, Shi et al.'s (2021) recommendations for a multidimensional assessment of urban resilience. Moreover, we have employed the FRAM to analyze the interrelationships among these resilience elements, thereby advancing the field.

The analysis of the FRAM model reveals that nodes with a k_s value of 4 account for 68 % of all nodes, highlighting the dynamic interactions between the indicators of urban resilience. Notably, the social security and employment expenditure indicator (F1) exhibits a significant out-degree, establishing its foundational role in supporting other resilience factors and corroborating Cochrane et al.'s (2023), findings on the impact of social welfare policies on urban resilience. This study further reveals that social security and employment expenditures, as direct

financial support from the government to citizens, not only elevate the per capita disposable income of urban residents (F8) but also increase the number of new jobs (F10) by creating job opportunities. This dual effect draws people into urban areas, contributing to the urbanization rate of the population (F4), and leads to increased public investment in healthcare, thereby expanding the number of medical institutions (F14). Additionally, this study found that the urbanization rate of the population (F4) indicator has a large in-degree, suggesting it is influenced by a multitude of factors. Enhanced levels of social security and employment expenditure (F1) and improved per capita area of roads (F12) act as magnets for urban migration, while growth in the number of new jobs (F10) and university students enrolled (F2) transforms this migration into sustained population growth, further amplifying the urbanization rate. These findings provide insights for population planning and development strategies in the Chengdu-Chongqing urban agglomeration.

This study extends beyond mere calculation and assessment of urban resilience to explore dynamic trends within the Chengdu-Chongqing urban agglomeration, addressing the previous limitation of confining resilience studies to static analyses (Ribeiro & Gonçalves, 2019; Datola et al., 2022; Schlör et al., 2018). Our case study reveals a distinctive two-headed structure in the resilience distribution, with Chengdu and Chongqing leading, while Ziyang, directly adjacent to both, significantly lags behind. Since 2016, a notable increase in resilience growth rates among other cities has been observed, demonstrating a trend of α -convergence that reduces the resilience gap. This trend aligns with findings from Lu, Zhang, et al. (2022), which are linked to a reassessment of urban vulnerability in the Sichuan Basin following the 2012 Wenchuan earthquake and subsequent resilience planning efforts. Additionally, spatial correlation analysis within the Chengdu-Chongqing urban agglomeration indicates a negative trend, suggesting that the advanced resilience of Chengdu and Chongqing may inhibit the resilience development of neighboring cities. This unique finding not only complements previous studies on the resilience of the Chengdu-Chongqing urban agglomeration (Lu, Zhang, et al., 2022; He et al., 2022) but also validates the “siphon effect” theory on regional resilience

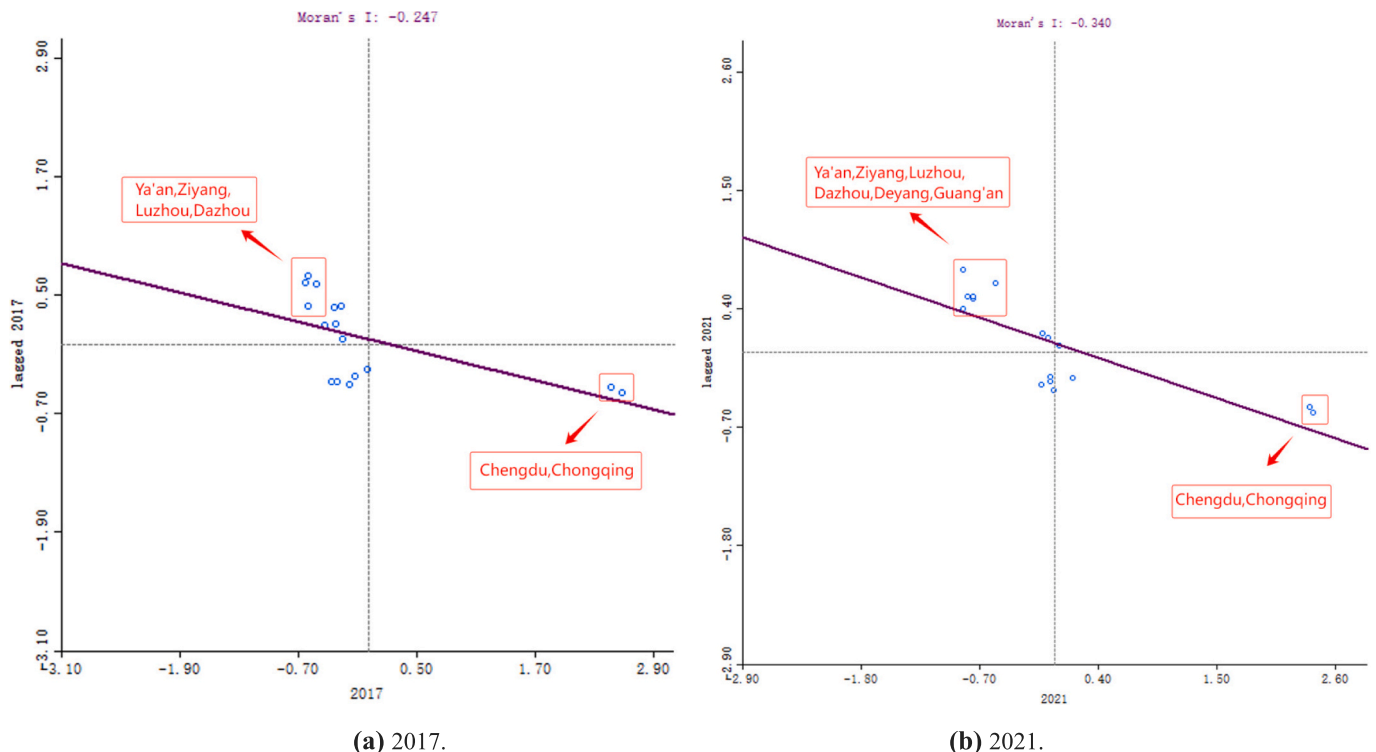


Fig. 17. Moran's I scatter plots for resilience in Chengdu-Chongqing urban agglomeration in 2017 and 2021.

Table 7
Test of absolute β -convergence spatial model for the resilience of Chengdu-Chongqing urban agglomeration.

OLM		SEM		SLM	
Variable	2012–2021	Variable	2012–2021	Variable	2012–2021
Constant	0.635	Constant	0.621	Constant	0.504
β	−0.134	β	−0.124	β	−0.124
s	0.013	λ	0.307	ρ	0.366
T	45.762	s	0.128	s	0.014
R^2	0.420	τ	53.271	τ	45.341
LM(LAG)	1.357 (0.133)	R^2	0.554	R^2	0.605
R-LM (LAG)	1.684 (0.141)	Log(L)	21.241	Log(L)	23.446
LM(ERR)	0.612 (0.375)				
R-LM(ERR)	0.477 (0.443)				

proposed by Wang and Zhang (2023). Furthermore, after accounting for spatial correlations, we observed a clear β -convergence trend, corroborating Wang and Yan's (2023) findings that cities with initially lower resilience levels are catching up faster, leading to a movement towards greater stability. Given these findings, policymakers must prioritize equitable resource distribution and develop targeted strategies to enhance the resilience of initially less resilient cities. It is essential to ensure that the development of core cities complements, rather than competes with, the development of neighboring regions. To this end, the implementation of coordinated and comprehensive urban development strategies, such as those exemplified by the 'Planning Outline for the Construction of the Chengdu-Chongqing Economic Circle' and the 'Development Plan for the Chengdu-Chongqing Urban Agglomeration', is critical.

6. Conclusion

Urban agglomerations have played a pivotal role in China's urbanization strategies, as reflected in the 11th, 12th, and 13th Five-Year Plans. This emphasis on urban clusters is further accentuated in the 14th Five-Year Plan. Here, the Chengdu-Chongqing urban agglomeration is projected to evolve into a significant new type of urban cluster in China within the next five years. Considering the importance of the Chengdu-Chongqing urban agglomeration, this study first constructs a quantitative assessment framework for city resilience from the perspective of complex adaptive systems, and comprehensively and objectively screens the resilience indicators through the systematic review method to ensure the transparency and traceability of the indicators. Secondly, this study innovatively proposes a data-driven quantitative assessment method for city resilience by integrating Functional Resonance Analysis Method (FRAM), modified K-shell (MKS) decomposition algorithm and Bayesian Networks (BNs) algorithm. Finally, the Chengdu-Chongqing urban agglomeration serves as a case study in this research. We assess the resilience levels of 16 cities within this region for the period 2012 to 2021. Employing spatio-temporal analysis, we delve into the evolutionary characteristics and patterns of city resilience, aiming to provide valuable references for the formulation

of resilient development strategies for relevant cities.

The results of this study illuminate key insights into the distribution and evolution of city resilience within the Chengdu-Chongqing urban agglomeration, as follows: (1) The distribution of city resilience in the Chengdu-Chongqing urban agglomeration exhibits a remarkable two-headed structure. In this structure, Chengdu and Chongqing are at the top, while Ziyang, which is directly adjacent to both, is at the bottom. (2) The differences in resilience between cities have gradually narrowed over time, showing a trend of α -convergence. (3) The Chengdu-Chongqing urban agglomeration shows negative spatial correlation. Chengdu and Chongqing with higher levels of resilience development limit the resilience development of neighboring cities to some extent. (4) Cities with initially lower levels of resilience show faster growth rates, leading to a gradual reduction of the differences in resilience levels between urban agglomerations and convergence to a steady state. Given these findings, the implementation of synergistic and integrated urban development strategies is crucial. These strategies are essential for mitigating negative spillover effects and ensuring that the development of core cities complements the progress of neighbouring regions, rather than competing with them.

Given the complexity of urban systems and the inherent limitations in data availability, this study's high data requirements have necessitated the selection of indicators that may not comprehensively cover all facets of urban resilience. Moreover, the lack of comprehensive data for the entire Chengdu-Chongqing urban agglomeration region has hindered a thorough assessment of its overall resilience. Despite these limitations, this study establishes a preliminary foundation for future research aimed at enhancing the resilience of urban agglomerations and contributes valuable insights into the field of urban resilience research.

CRediT authorship contribution statement

Zihao Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhijian Li:** Writing – review & editing, Visualization, Methodology, Investigation, Conceptualization. **Cong Lu:** Visualization, Methodology, Investigation, Data curation. **Jianjun She:** Supervision, Project administration, Funding acquisition. **Yilun Zhou:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Adjacency matrix of the complex network based on the FRAM model

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25
F1	0	1	0	1	1	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	1	1	1	1
F2	1	0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F3	0	0	0	0	1	0	0	0	0	1	0	1	0	1	0	0	0	1	0	1	1	0	0	0	0
F4	1	1	0	0	0	1	0	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0
F5	1	1	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0
F6	0	1	0	1	0	0	0	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1
F7	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1
F8	1	1	0	0	0	0	0	0	0	0	1	0	0	0	1	1	0	0	0	0	0	0	0	0	0
F9	1	0	0	0	0	1	0	0	0	0	1	0	1	1	0	0	1	0	0	0	0	0	0	0	1
F10	1	0	1	1	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F11	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F12	0	0	1	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1
F13	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0
F14	1	0	1	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
F15	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
F16	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0
F17	0	0	0	1	0	0	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0	1	0	0
F18	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0	1	0	0
F19	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	1
F20	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
F21	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1
F22	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0
F23	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0	0	0
F24	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
F25	1	0	0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	1	0	1	0	0	1	0

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cities.2024.105226>.

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